

R. Divika Rathi

Jakarta, Indonesia

rdivika.rathi@binus.ac.id | [linkedin.com/in/divika-rathi](https://www.linkedin.com/in/divika-rathi) | github.com/divikazz | [Portfolio](#)

Computer Science student at BINUS University with interests in data analytics, databases, data engineering, and digital products. Experienced in SQL, Python, ETL workflows, data visualization, and machine learning through academic and team projects, alongside UI/UX design for product concepts in Figma. Interested in applying technical and analytical skills to solve business problems and communicate data-driven insights.

Education

BINUS University

(2024 - Present)

Bachelor of Computer Science - GPA 3.33/4.00

Experience

HIMTI BINUS@Bandung

(2024 - Present)

Regional Secretary

- Coordinated documentation, correspondence, schedules, and administrative workflows across organizational activities.
- Worked with multiple teams to organize event requirements and keep communication clear from planning through execution.

BINUS University (Freshmen Program)

(Sep 2025 - Sep 2026)

Freshmen Partner

- Supported freshmen through their transition into university life with guidance, information, and assistance.
- Coordinated communication with fellow partners to support a positive onboarding experience.

BINUS University (Freshmen Program)

(Jul 2025 - Sep 2025)

Freshmen Leader

- Guided incoming freshmen during the university transition period and supported program activities.
- Worked with student leaders to share information and assist freshmen throughout the program.

Projects

Online Retail Data Engineering & Business Intelligence (Python, Pentaho PDI, MySQL, Power BI)

- Built an end-to-end ETL pipeline for 541,909 Online Retail transactions, including preprocessing, duplicate and missing-data handling, and data warehouse loading.
- Designed a star schema with product, customer, time, and sales fact tables and developed Power BI dashboards for sales, customer behavior, and product performance analysis.

Home Credit Default Risk: Big Data Processing (PySpark, Hadoop HDFS, Random Forest)

- Processed Home Credit loan data stored in HDFS with PySpark, covering cleaning, validation, aggregation, and feature engineering such as credit-to-income ratios.
- Trained a Random Forest classifier to predict loan default, reaching an AUC of 0.74.

Linear Regression & Greedy Join Order Optimization (SQL, MySQL, Python, Scikit-learn)

- Built a query-cost prediction framework using 720 MySQL query executions and 17 structural features, training Linear Regression on log-transformed execution times.
- Applied a Greedy Algorithm to select lower-cost join orders; achieved R^2 87.45%, RMSE 0.3044 ms, and MAE 0.2206 ms across 128 join-order scenarios.

Terrascent (Figma, UI/UX, Business Model)

- Group project developing an eco-friendly perfume brand concept with reusable wooden bottles, a scent-matching quiz, and a return-bottle loyalty system.
- Deliverables include user personas, an empathy map, a perceptual map, revenue and cost structure, and a Figma mobile prototype covering nine app flows.

Buylo (Figma, UI/UX, HCI)

- Designed a mobile e-commerce application concept around a fast, convenient shopping experience, covering user-centered design, Figma prototyping, and usability evaluation through user testing.

Isyaratku (Django, FastAPI, SQLite, AI, UX/UI)

- Developed a sign language learning platform combining interactive lessons with AI-assisted gesture recognition.
- Designed user flows for the dictionary, gesture practice, quizzes, and progress tracking and contributed to backend and UI integration.

SightSense (AI, Product Design)

- Competition project (Samsung Solve for Tomorrow 2025, preliminary round) exploring AI and smartphone technology for more accessible early cataract screening.
- Designed the screening-result and educational experience and collaborated on the product concept and presentation.

Certifications

Oracle Dev Gym (Certificate of Excellence)

Databases for Developers: Foundations - Grade 100%

IBM / Cognitive Class

(Feb 2025)

Python 101 for Data Science (PY0101EN)

Skills

Programming: Python, SQL, JavaScript, C, Java

Data: Pandas, PySpark, Scikit-learn, BigQuery, Power BI, Tableau

Database and ETL: MySQL, Pentaho Data Integration, Hadoop HDFS

Development: Next.js, Django, FastAPI

Design: Figma, UI/UX, Prototyping

Tools: GitHub, Google Colab